५ क्षी ग्णेशायनम :५

Featurization & Feature Engineering

↓

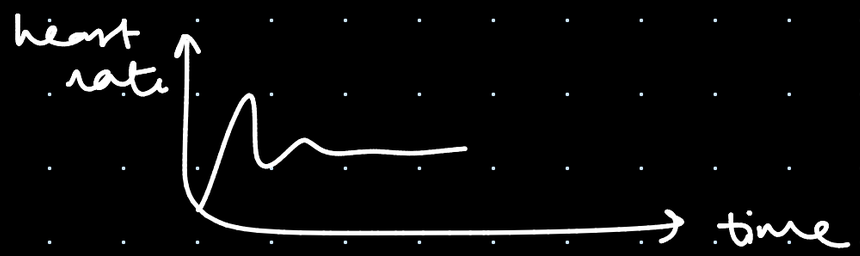
FE

For text data : Features of any data → numerical data.

E.g. text data : Bag of words, tf-idf, avgw2v

categorical data : one-hot encoding, mean response rate

Time series data :

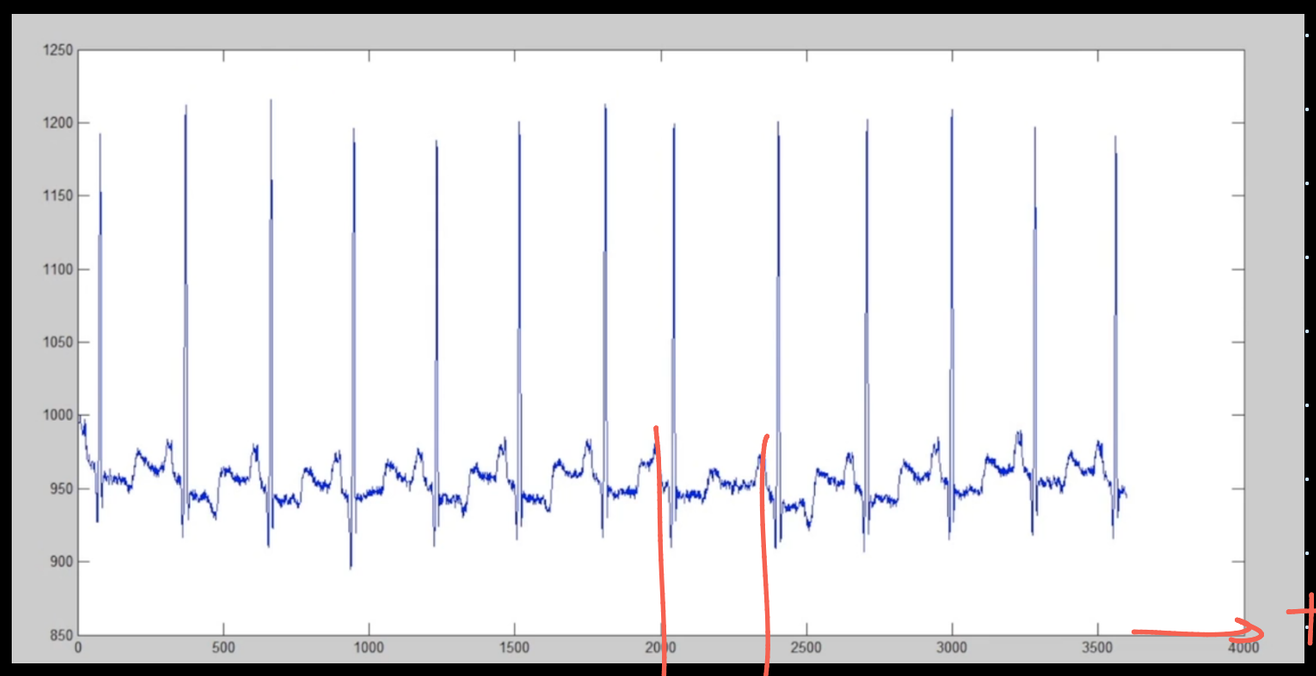


Audio is repr. as time series data.

text can also be thought of as a sequence.

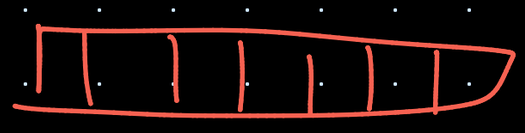
Moving Window for Time Series

↳ simplest one for TS



← width
Take a window (snapshot of time)

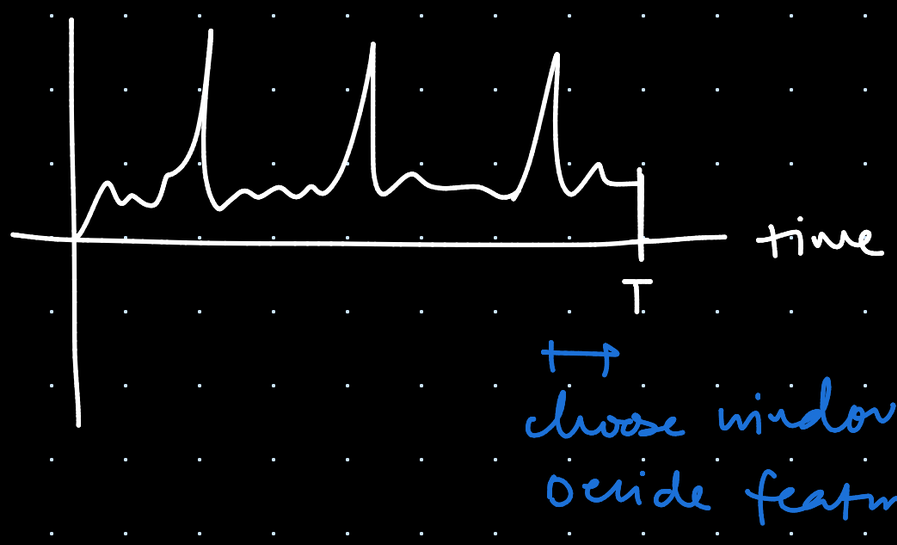
Take data from window.



put each val in vector

① Calc mean, std dev, etc.
② medians, quantiles
③ max, min
④ max-min, max/min
⑤ # of local minima/maxima
⑥ zero-crossing / mean-crossing

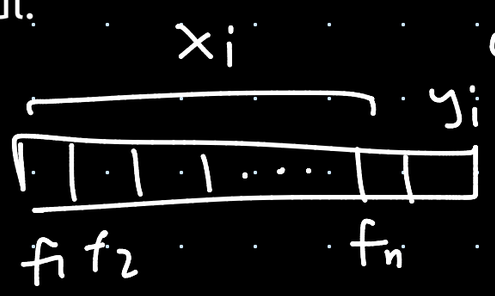
} all of these can be treated as features (dep. on domain expert.)



← choose window
decide feature

Say task: predict cardiac arrest in next 10 mins

1. Define window width.
2. Check features that are useful.
3. Make vector out of features.



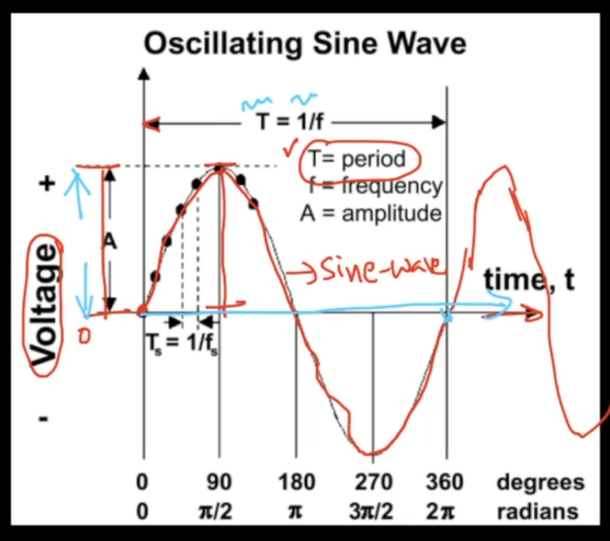
← pt. gets cardiac arrest / not.

Fourier decomposition/Transform

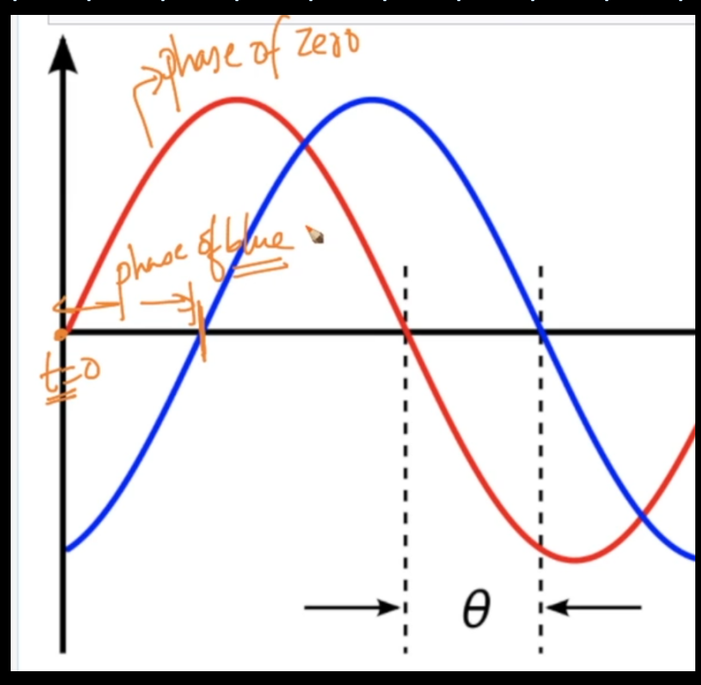
→ to represent time series

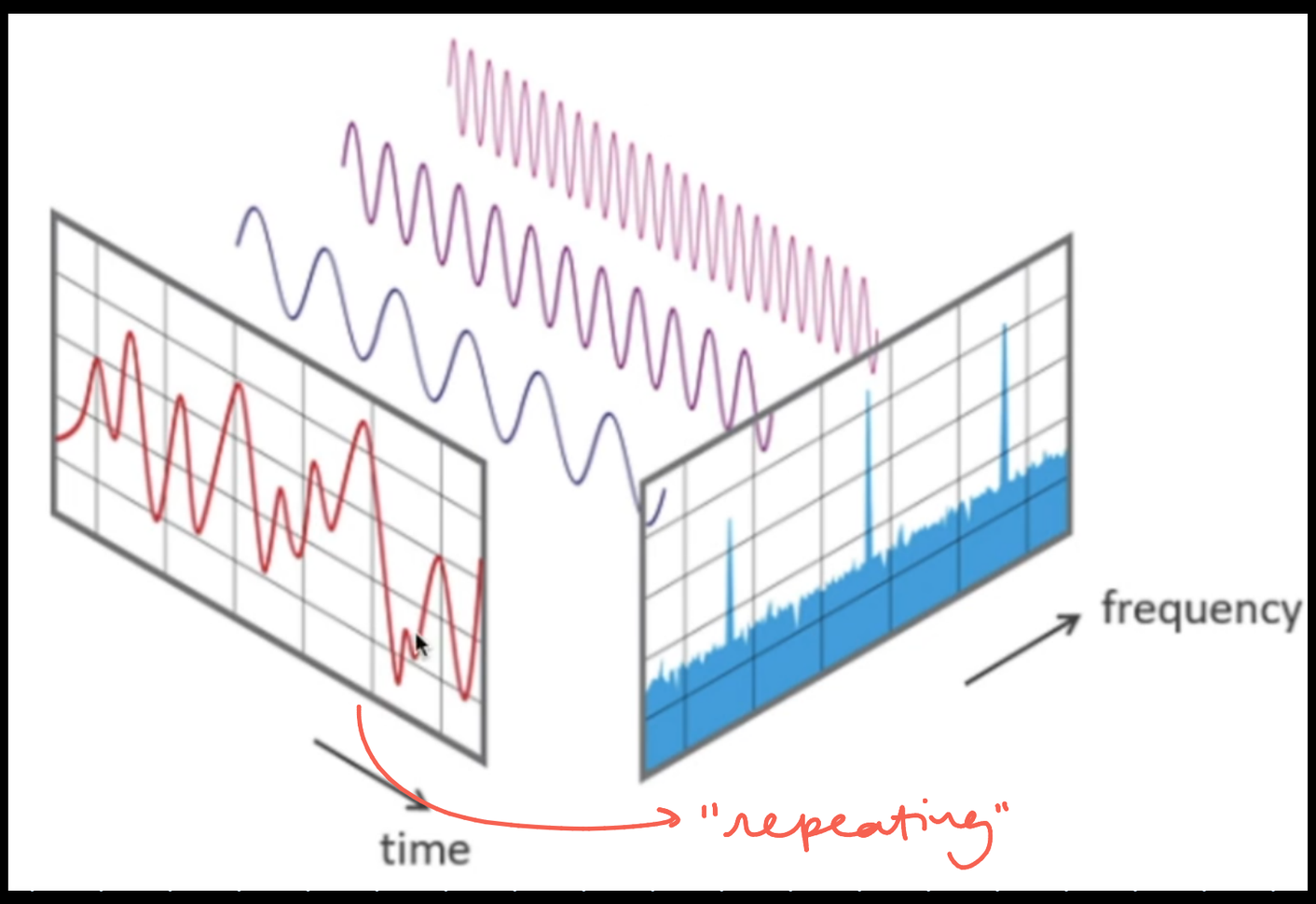
one osci per second = 1 Hz

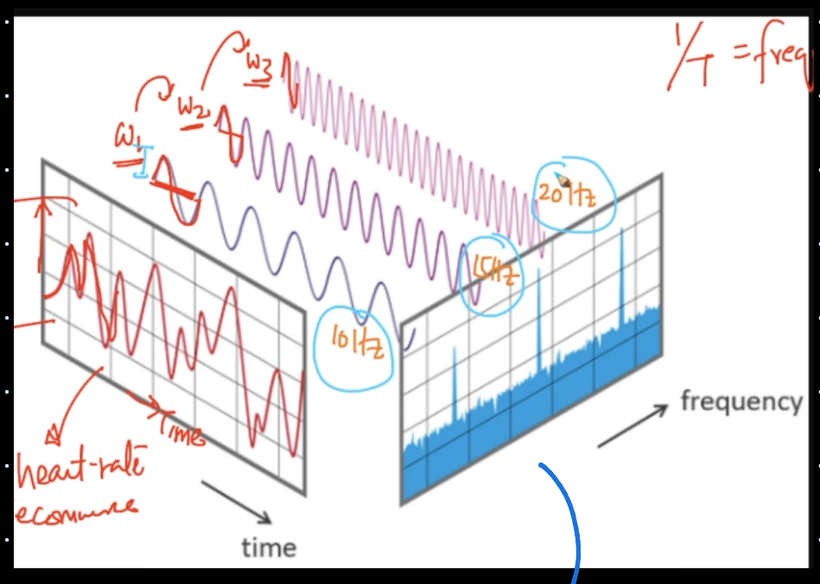
2 osci per sec = 2 Hz



amplitude
Time period
freq = 1/T





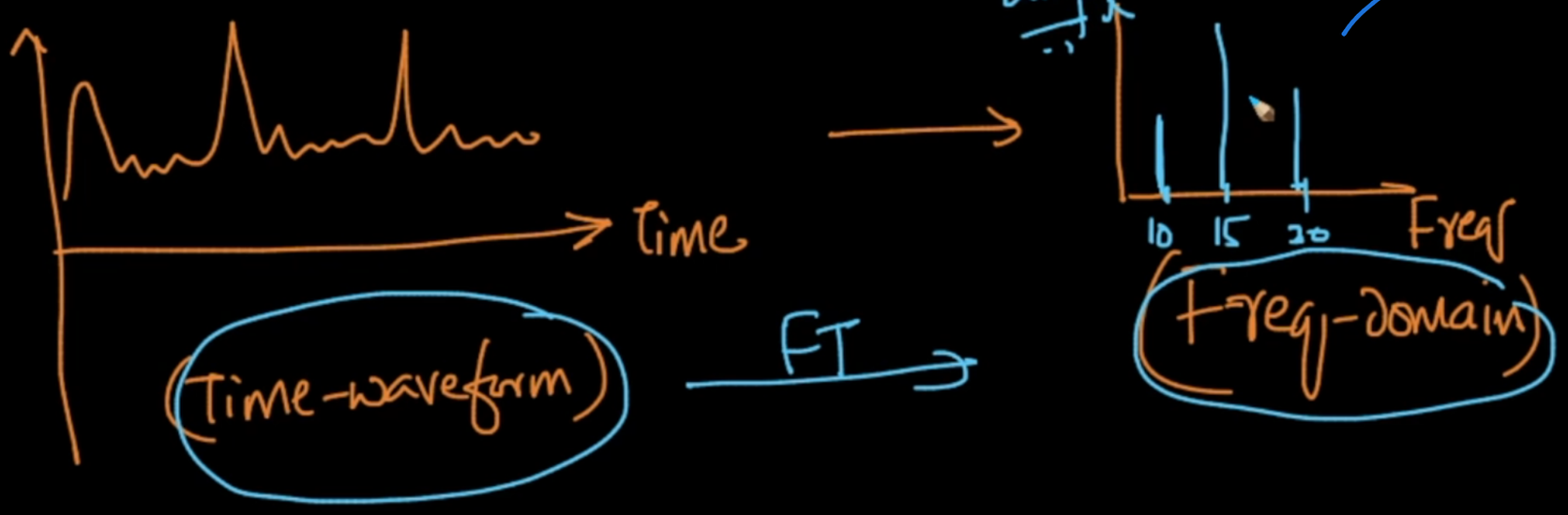


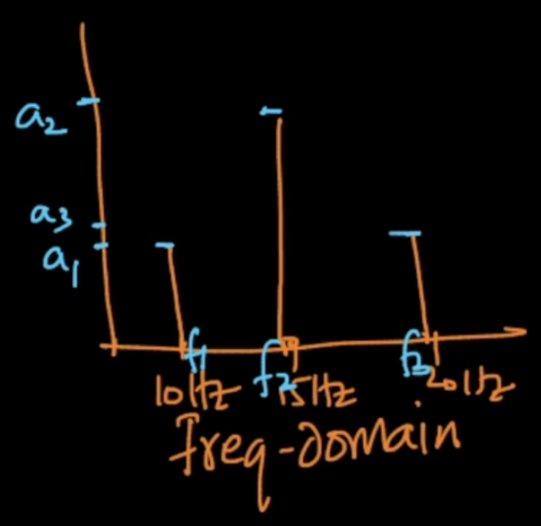
The idea is that given any composite wave, (repeating) freq-domain
we can break it into a sum of diff sine waves.

↓

at diff frequencies

→ useful esp. for "repeating" patterns.





Step 1: using FT, get diff. frequencies and their corresponding domains.

Say f_1, a_1 f_2, a_2 f_3, a_3 are the significant ones.

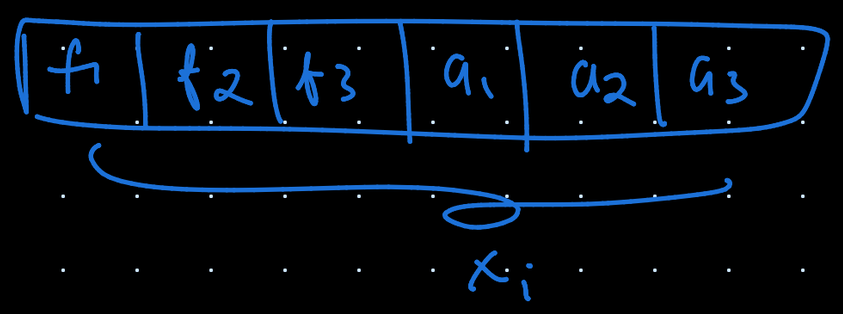
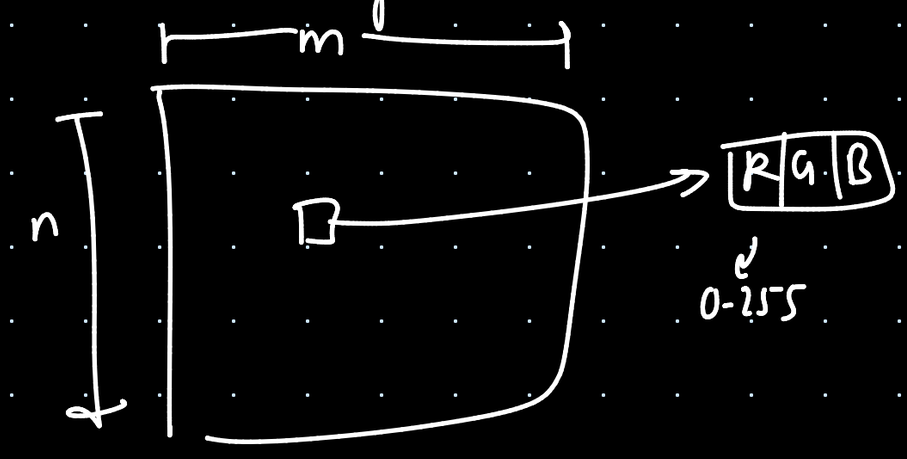


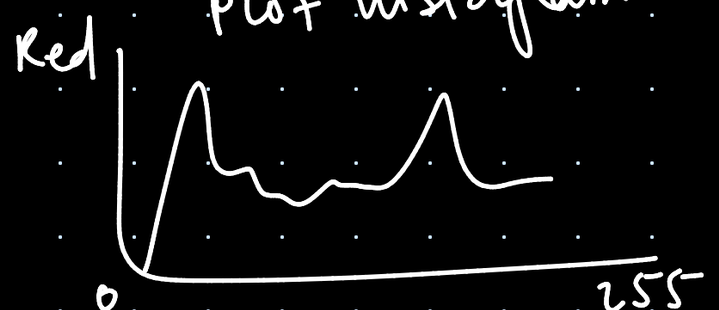
Image Histograms

→ color histogram
→ edge histogram

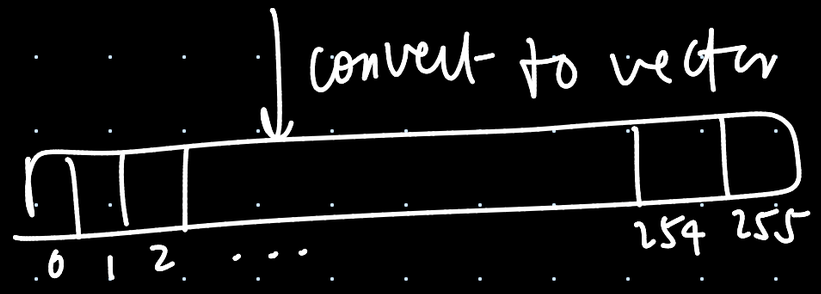
color histogram.



take all red values for each pixel.
n x m datapoints
plot histogram:



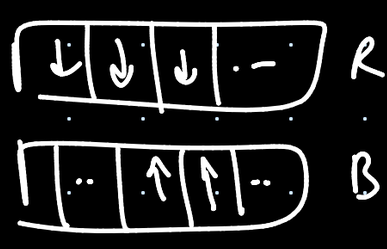
convert to vector



put in frequencies

For all colors:  R G B vectors.

So say to check if an image is of the sky:



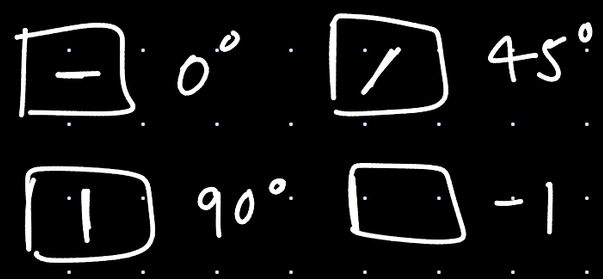
we could look for particular types of comb^ns.

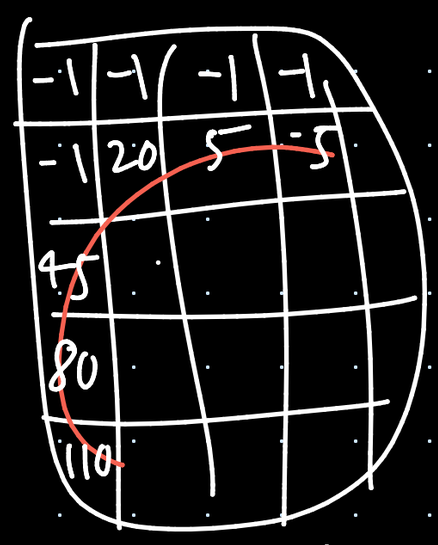


Obv. this wouldn't do really complex stuff:
It can't recognize shapes.

Edge histogram:



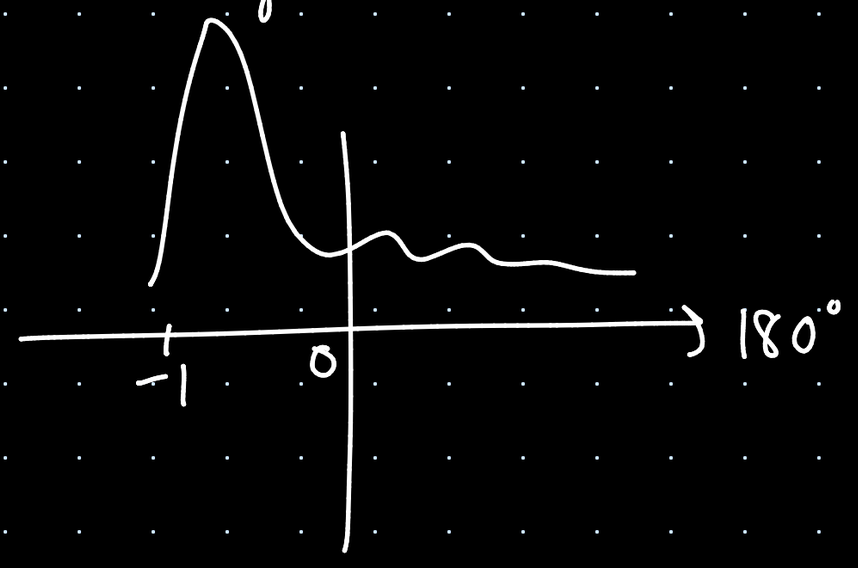




grid of vals.

So for every loca^n, we can get an angle value.

Now we can plot histogram.



plot freq. for each angle.

Histograms are rudimentary.

Scale Invariant Feature Transforms (SIFTs)

(s used for obj. detec^n.



$I = \{V_1, V_2, \ldots, V_{10}\}$

128-dim 128-dim

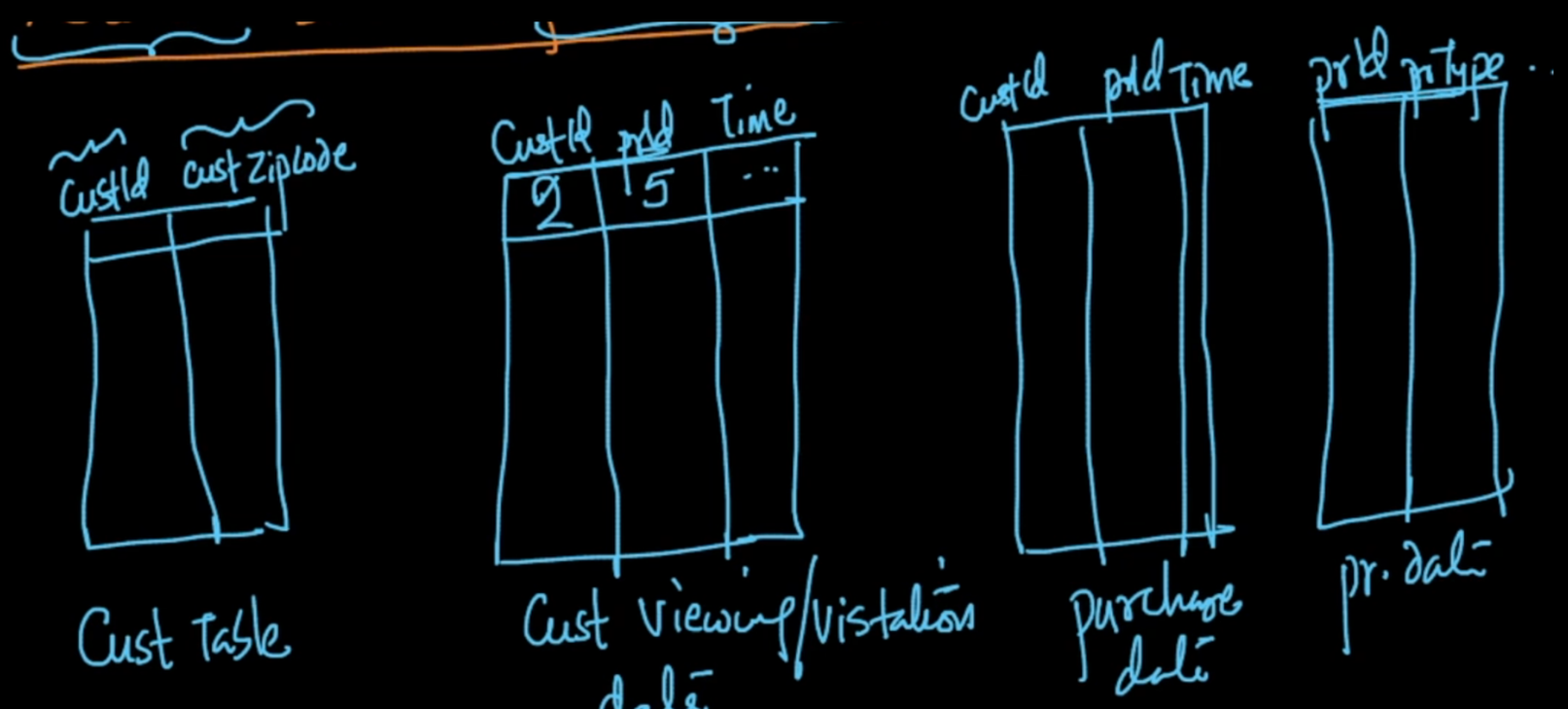
For every keypoint, create 128-dim vec.

Has scale and rotation invariance.

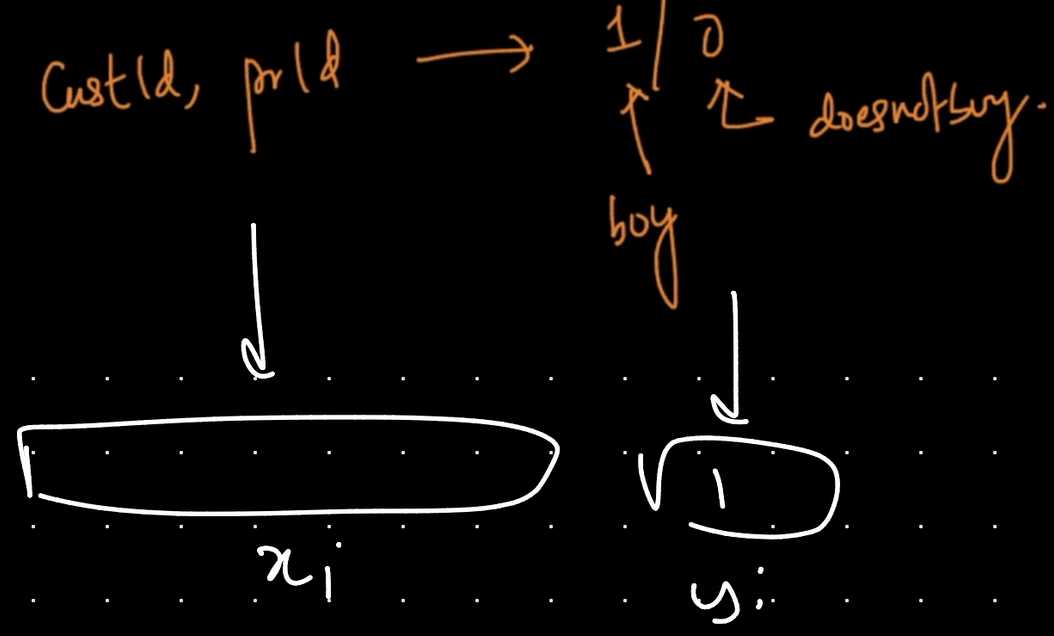
OpenCV has SIFT.

Relational Data

consider this example

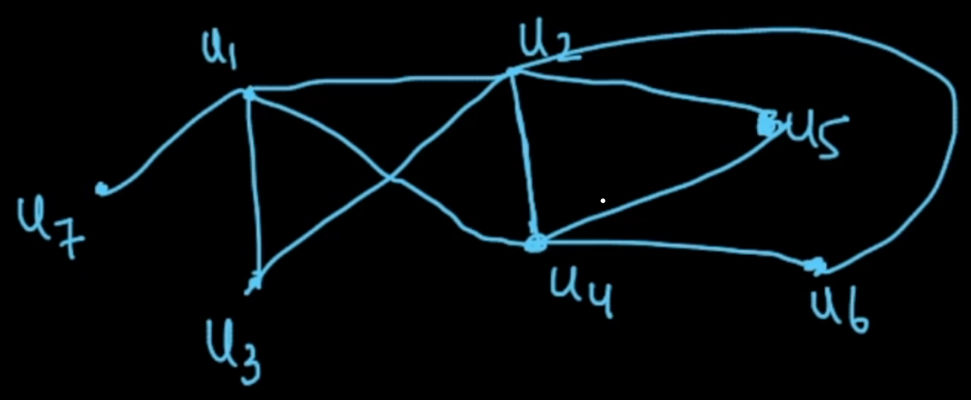


Task: predict if a cust would purchase a product in the next 7 day



choose features based on domain expertise

Graph Data



↙ Consider this social graph.

Task is to recommend new friends for u_i

given u_i, $u_1 \longrightarrow$ 1/0 if friends or not.

Say for rec. to u_4.

check $u_4 - u_3 \longrightarrow$ 2 mutuals (u_1, u_2)

$u_4 - u_7 \longrightarrow$ 1 mutual (u_1)

f_1 (# of mutuals)	f_2: # of paths betn vertices

↙ get these using graph theory algos.

Feature Engg. (Indicator variable)

Say height is a feature.

↳ can leave as real-valued

or use indicator $>150 \rightarrow 1$

$\leq 150 \rightarrow 0$ } Binary

Feature Binning / Bucketing

extension of indicators.

$h < 120 \rightarrow 1$

$h < 150$ and $\geq 120 \rightarrow 2$

$h < 180$ and $h \geq 150 \rightarrow 3$

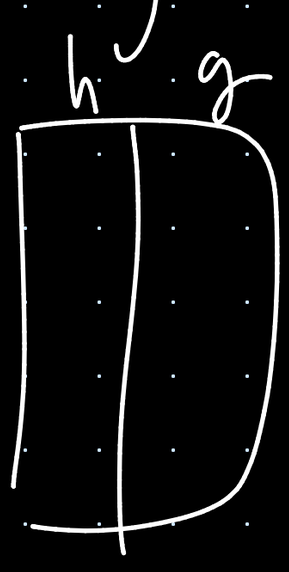
$h \geq 180 \rightarrow 4$ } categorical

Thresholds are problem specific.

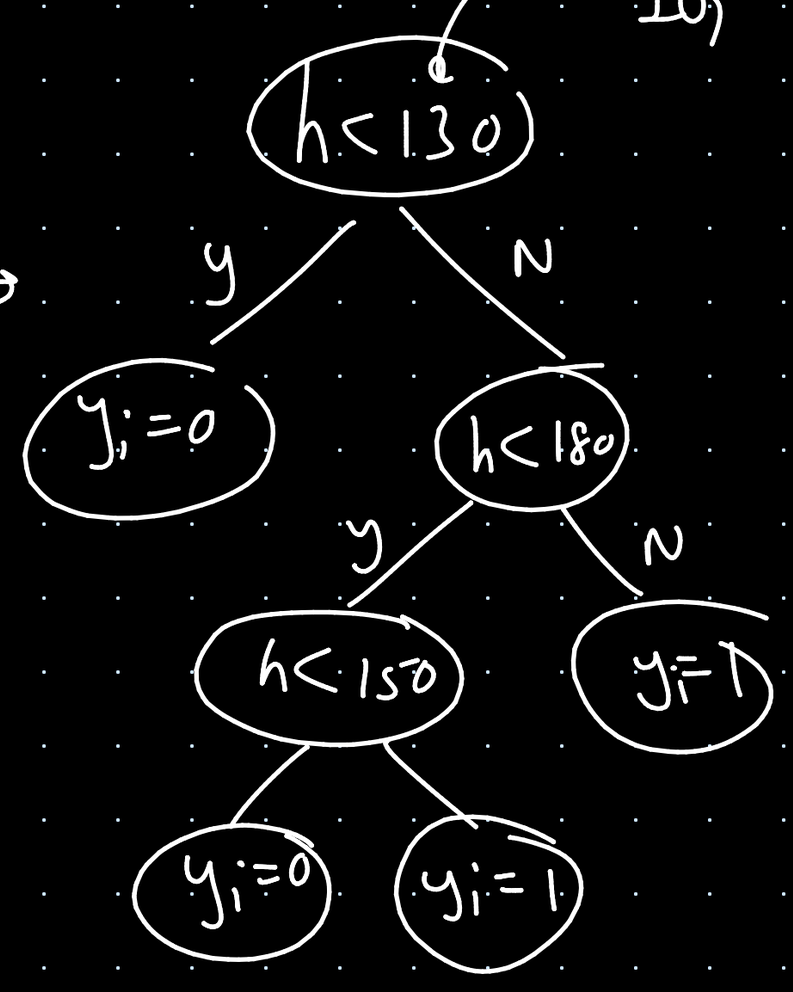
Say we want to bin h to predict gender.

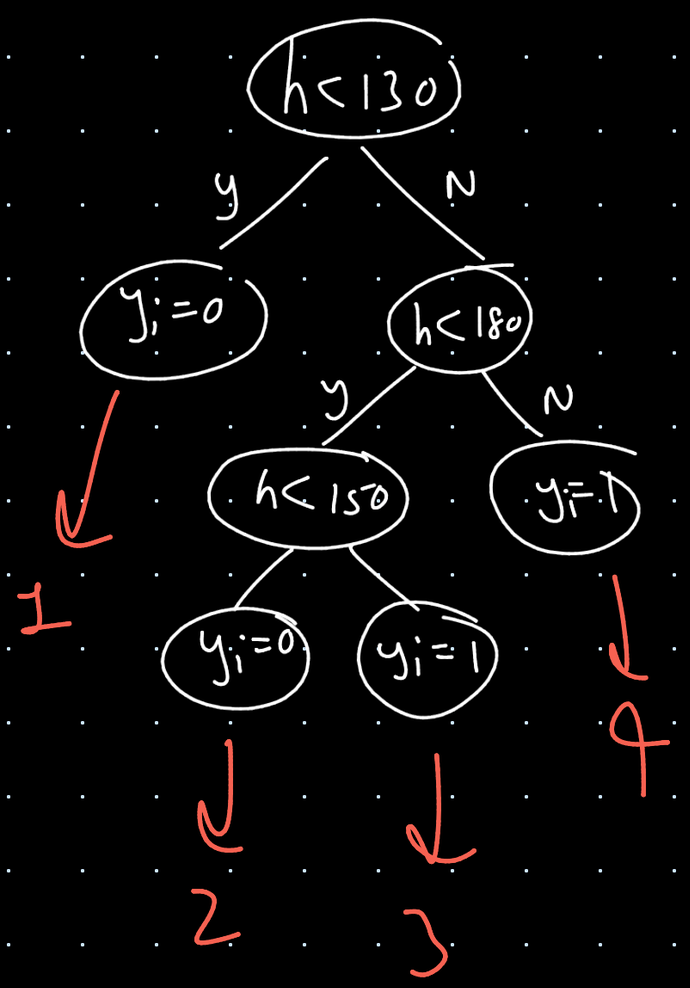
Data-driven binning:

Soln: make table



simple DT →





category for every leaf.

Interacⁿ variables

x_i: {h, w, hL, ec} y_i: gender

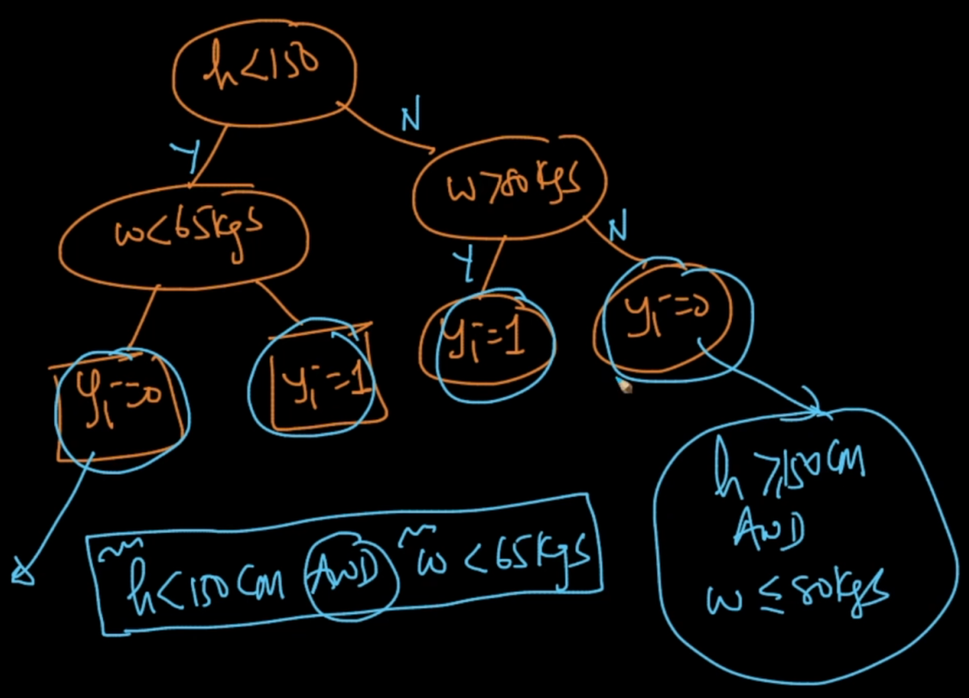
ht. wt. hair len. eye colour

e.g. ① (h < 150cm) AND (w < 60 kgs) } logical 2-way interacⁿ.

② h * w } mathematical 2-way interacⁿ feature.

Q- How do you find good interacⁿ features!

→ DTs.



Mathematical Transforms

$x \rightarrow$ single feature

can apply operaⁿs like $\log(x)$; e^x; $\sin x$; $\sqrt{x}$; x^2; ...

what is the best transform? problem-specific

e.g. If x has a power law distribution.

$\log x$ would make sense.

It'd be Gaussian.

model-specific featurizations

e.g. Say f_1 is power-law distributed.

And if we wanted to use log. reg^n or GNB,
they assume that features are Gaussian dist.
So $f_1 \to \log f_1$ would be helpful.

e.g. BoW usually gives high-dim spaces.
So linear models work okay: (lin. SVM, logis. reg^n)
DTs may not work well

Feature Orthogonality

* the more orthogonal the features are, the better would the model be.

(Idea) $f_1, f_2, f_3 \xrightarrow{(M)} y_i$

$M :- \tilde{y}_i$



(Q) how do I design a new feature f_4 s.t

$f_4 \xrightarrow{\text{corr}} y_i$

less corr with f_1, f_2, f_3

↳ we already have a model.
we want a new f_4 that has corr^n with y but not with the other features.

error : $e_i = y_i - \tilde{y}_i$

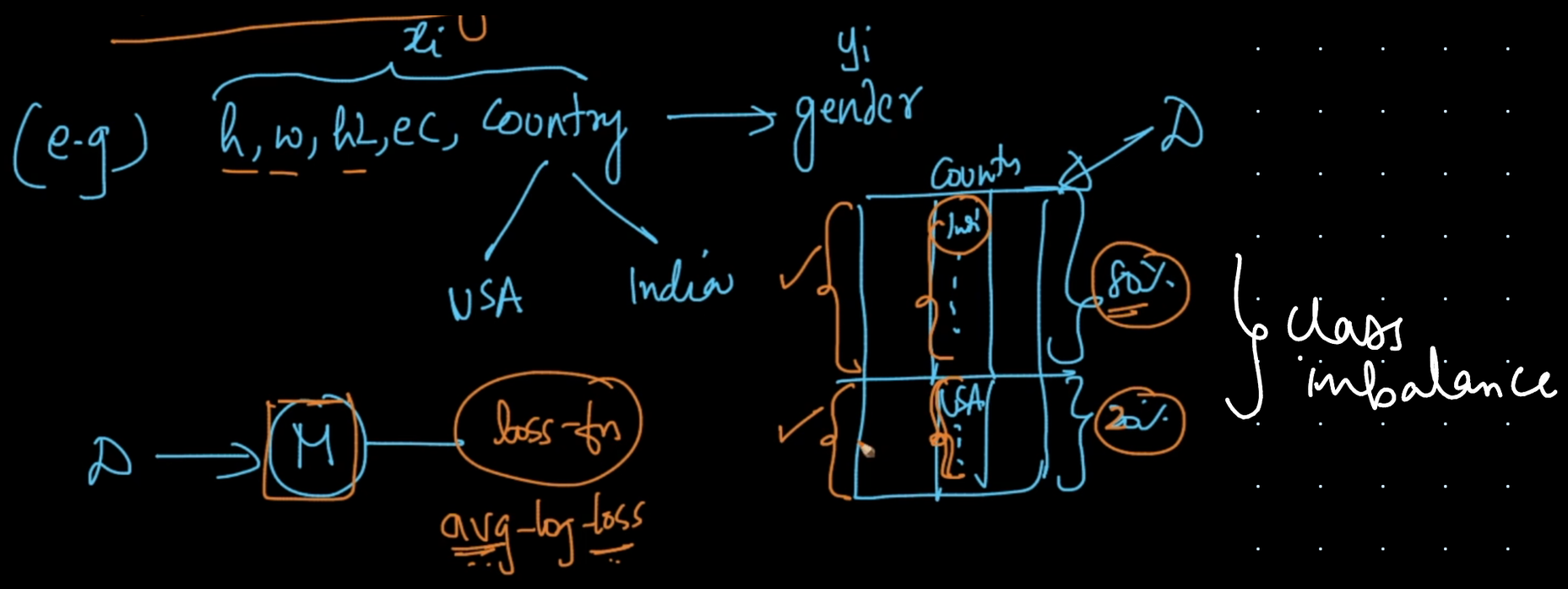
make sure $f_4 \xrightarrow{\text{corr}} e_i$

M' trained on f_1, f_2, f_3, f_4 would work better.

Idea is similar to boosting (GBDT).

But it is easy to overfit over here.

Feature Slicing



When there's a distinct bet" values of one feature, the model will try to perform better on 1 val.

Here, India is the country for 80% samples.

So model tends to perform better on Indians.

worse for US points.

Strategy
- D_{India} train M_1
- D_{US} M_2

Requirement:

① Categories 1 & 2 are quite diff. in their behavior.

② Sufficient data pts. for both slices.

e.g. Desktop & mobile traffic